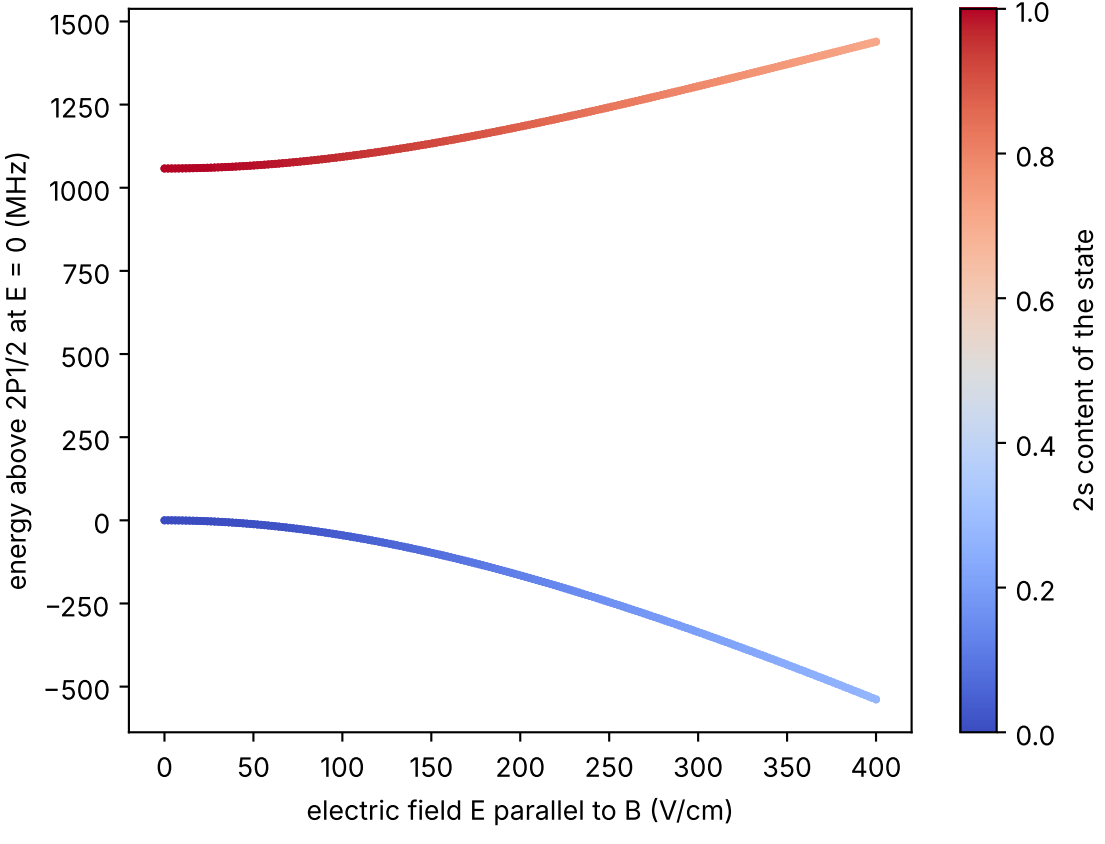
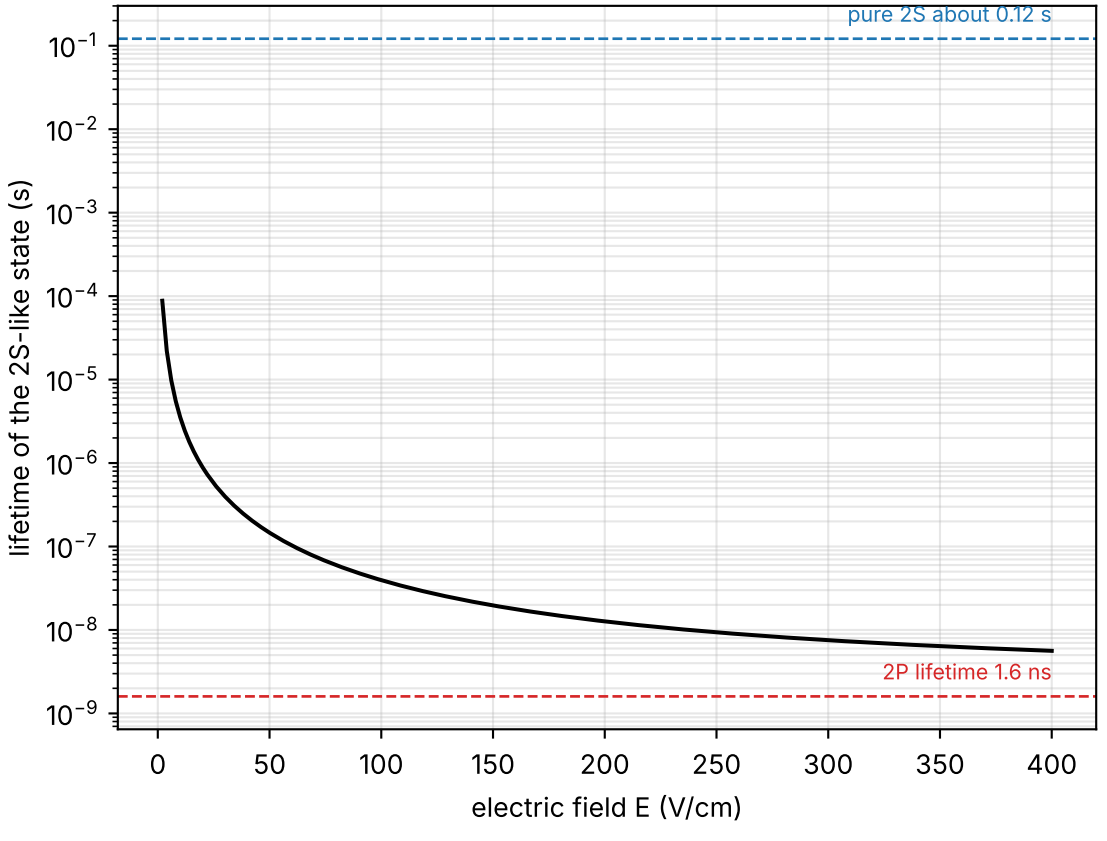

2S1/2 and 2P1/2 repel and mix as E grows
energy above 2P1/2 at E = 0 (MHz)
1500
1250
1000
750
500
250
0
−250
−500
0
50
100
150
200
250
300
350
400
electric field E parallel to B (V/cm)
1.0
0.8
0.6
0.4
0.2
0.0
2s content of the state


Quenching of the metastable 2S state
lifetime of the 2S-like state (s)
$10^{-1}$
$10^{-2}$
$10^{-3}$
$10^{-4}$
$10^{-5}$
$10^{-6}$
$10^{-7}$
$10^{-8}$
$10^{-9}$
pure 2S about 0.12 s
2P lifetime 1.6 ns
0
50
100
150
200
250
300
350
400
electric field E (V/cm)